

Electronic Circuits I

Analog Harmonic Tremolo

Design, Construction, and Experimental Validation

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Contents

1	Introduction	1
2	Control Circuit	1
2.1	Signal Generator	2
2.2	Phase Splitter	2
2.3	Design Criteria	3
3	Audio Circuit	5
3.1	Buffer Stages and Biasing	5
3.2	Filters	6
3.3	Optical Gain Control	7
3.4	Measured Attenuation	8
4	Prototype Specifications	9
5	Complete Circuit	9
5.1	Control Circuit	9
5.2	Audio Circuit	10
6	Conclusions	11
A	Audio-Circuit Capacitor Calculations	11
A.1	Low-Pass Circuit: Capacitors with a 5 Hz Cutoff	12
A.2	Low-Pass Filter Capacitor for an 800 Hz Target Cutoff	14
A.3	High-Pass Circuit: Capacitors with a 5 Hz Cutoff	15
A.4	High-Pass Filter Capacitor for an 800 Hz Target Cutoff	17

1 Introduction

The aim of this project was to design and build an analog harmonic tremolo for electric guitar, powered by a single 9 V supply. Unlike a conventional tremolo, which modulates the level of the entire signal uniformly, a harmonic tremolo splits the audio signal into two frequency bands and modulates their amplitudes in opposite phase. As the level of the low-frequency band increases, the level of the high-frequency band decreases, and vice versa. This alternating behavior produces a periodic variation in timbre in addition to amplitude modulation.

The system was organized into two main sections:

1. A control circuit consisting of a low-frequency oscillator (LFO), a depth control stage, a gain stage, and a phase splitter that generates two opposing control signals.
2. An audio circuit that divides the guitar signal into low-pass and high-pass branches, controls the attenuation of each band through LED–LDR optocouplers, and finally recombines both branches.

The tremolo rate control adjusts the LFO frequency between approximately 1 Hz and 10 Hz. The LFO output, with an amplitude of approximately 500 mV peak, passes through a depth control stage and a gain stage with a nominal gain of approximately 12 dB before reaching the phase splitter. The depth control attenuates the LFO signal from its maximum amplitude down to zero without altering the DC operating point of the LEDs.

Both audio branches were designed around a nominal crossover frequency of 800 Hz. During implementation, standard component values were used and some components were adjusted experimentally. As a result, the assembled low-pass branch exhibited a cutoff frequency of approximately 1.2 kHz, while the high-pass branch remained close to 800 Hz. This discrepancy is discussed as a limitation of the prototype, but it did not prevent the operating principle of the effect from being successfully verified.

The prototype was built, tested using laboratory instrumentation, and publicly presented at CAPER 2024 and WorkShow 2024. Figure 1 shows the overall system architecture.

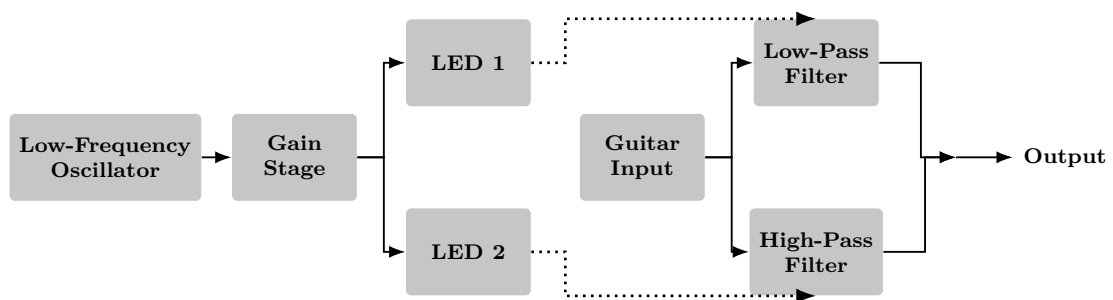


Figure 1: Block diagram of the harmonic tremolo.

2 Control Circuit

The control circuit generates two low-frequency signals in opposite phase. It consists of a low-frequency oscillator (LFO), a passive depth control, a gain stage, a phase splitter, and two buffer stages that control the LED currents.

2.1 Signal Generator

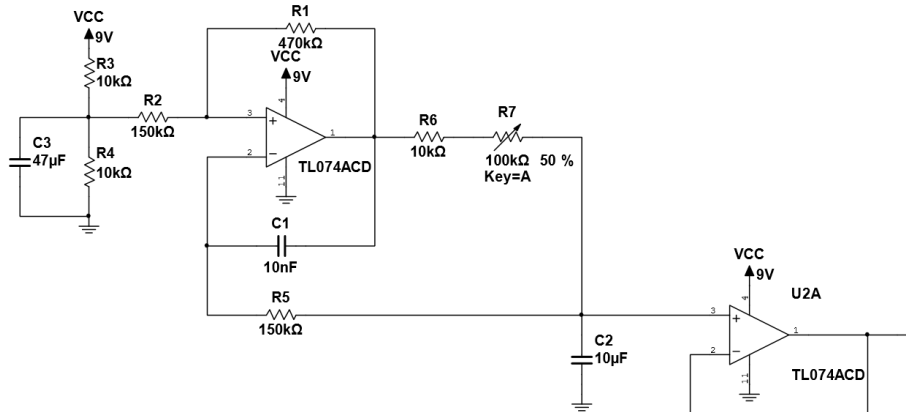


Figure 2: Low-frequency oscillator (LFO).

The oscillator topology was adopted as a starting point and was not designed as part of this project. The following characteristics were used to design the subsequent stages:

- output frequency adjustable between approximately 1 Hz and 10 Hz;
- output amplitude of approximately 500 mV peak;
- output impedance considered negligible relative to the following stage due to the operational-amplifier voltage follower.

The depth control potentiometer is placed before the gain stage and allows the AC control signal to be reduced to nearly zero without altering the DC operating point of the LEDs.

2.2 Phase Splitter

The phase splitter duplicates the control signal and inverts one of its outputs. As a result, when the current through one LED increases, the current through the other decreases, causing the two audio bands to be modulated in opposite phase.

The AC control signals are coupled to the LED buffer stages through capacitors. This allows the same DC bias point to be established experimentally at both transistor bases while the opposing AC signals from the phase splitter are superimposed on that operating point.

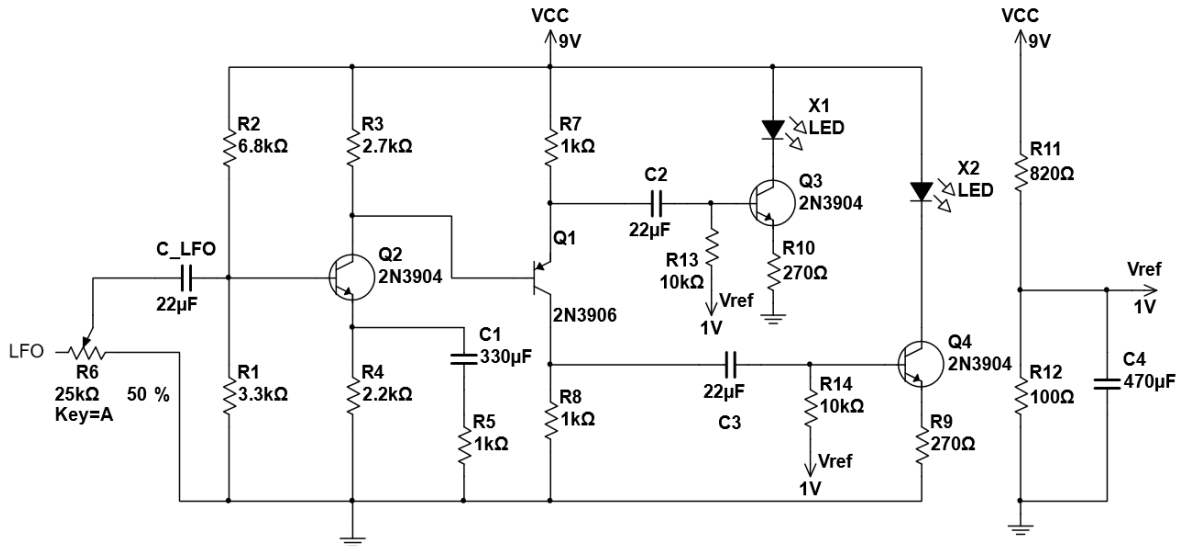


Figure 3: Gain stage, phase splitter, and LED buffer stages.

2.3 Design Criteria

2.3.1 Depth Control

The depth control was implemented using a potentiometer configured as an attenuator. At one end of its range, it delivers the maximum control signal; at the other, it reduces the oscillating component to nearly zero. In this condition, the LEDs remain at their fixed DC bias point and the tremolo effect disappears.

2.3.2 Phase-Splitter Biasing

The main design criterion was to obtain an approximately symmetrical output swing. The selected nominal operating point was:

- $V_{R_E} = 2.25 \text{ V}$;
- $V_{CEQ} = 4.5 \text{ V}$;
- $V_{R_C} = 2.25 \text{ V}$.

With $1\text{ k}\Omega$ collector and emitter resistors, the resulting DC bias current is

$$I_{CQ} = \frac{2.25 \text{ V}}{1 \text{ k}\Omega} = 2.25 \text{ mA}. \quad (1)$$

Assuming $V_{BE} \approx 0.65$ V, the nominal base voltage is approximately 6.1 V. Considering the 10 k Ω load presented by the following stage, the estimated output-swing limits are

$$V_{o,\max}^{(+)} \approx \frac{V_{CEQ}}{2} = 2.25 \text{ V}, \quad (2)$$

$$V_{o,\max}^{(-)} \approx I_{CQ} (1 \text{ k}\Omega \parallel 10 \text{ k}\Omega) = 2.05 \text{ V}. \quad (3)$$

The difference between these values is small relative to the required output amplitude of approximately 2 V peak.

2.3.3 Gain Stage

The gain stage was designed to amplify the maximum 500 mV peak output of the LFO to approximately 2 V peak. The target voltage gain was 4, equivalent to 12.0 dB.

The voltage available across R_C is

$$V_{R_C} = 9\text{ V} - 6.1\text{ V} = 2.9\text{ V}. \quad (4)$$

For $R_C = 2.7\text{ k}\Omega$, the resulting DC bias current is

$$I_{CQ} = \frac{2.9\text{ V}}{2.7\text{ k}\Omega} = 1.07\text{ mA}. \quad (5)$$

With $R_E = 2.2\text{ k}\Omega$, the resulting voltage drops are $V_{R_E} \approx 2.35\text{ V}$ and $V_{CEQ} \approx 3.75\text{ V}$. Using $h_{fe} = 120$ and $h_{ie} = 3.6\text{ k}\Omega$, the calculated voltage gain is 3.76, or

$$20 \log_{10}(3.76) = 11.5\text{ dB}. \quad (6)$$

This value is sufficiently close to the nominal target of 12 dB.

2.3.4 LED Biasing

The LEDs were biased at an intermediate operating point so that the AC control signal could both increase and decrease their brightness throughout the complete modulation cycle. A reference voltage of approximately 1 V was established experimentally. The selected voltage divider provides

$$V_{ref} = 9\text{ V} \frac{100\ \Omega}{100\ \Omega + 820\ \Omega} = 0.98\text{ V}. \quad (7)$$

With a $270\ \Omega$ emitter resistor and $V_{BE} \approx 0.65\text{ V}$, the approximate DC bias current of each transistor is

$$I_{CQ} = \frac{1\text{ V} - 0.65\text{ V}}{270\ \Omega} = 1.30\text{ mA}. \quad (8)$$

This value remains well below the 10 mA limit adopted for the LEDs. The $10\text{ k}\Omega$ base resistor limits the load presented to the phase splitter.

2.3.5 Coupling Capacitors

The capacitors in the control circuit were initially designed for a cutoff frequency of 0.5 Hz, below the minimum LFO frequency.

For the LFO output coupling capacitor, the estimated equivalent resistance was

$$R_{C_{LFO}} = 25 \text{ k}\Omega + 6.8 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega \parallel [h_{ie} + h_{fe} (2.2 \text{ k}\Omega \parallel 1 \text{ k}\Omega)] = 27.17 \text{ k}\Omega. \quad (9)$$

Therefore,

$$C_{LFO} = \frac{1}{2\pi(0.5 \text{ Hz})(27.17 \text{ k}\Omega)} = 11.7 \text{ }\mu\text{F}. \quad (10)$$

The capacitor value was experimentally increased to $22 \text{ }\mu\text{F}$ to reduce the attenuation observed between 1 Hz and 3 Hz.

For the gain-stage bypass capacitor, the estimated equivalent resistance was approximately $1.046 \text{ k}\Omega$. The ideal capacitance for a cutoff frequency of 0.5 Hz is

$$C = \frac{1}{2\pi(0.5 \text{ Hz})(1.046 \text{ k}\Omega)} = 304 \text{ }\mu\text{F}. \quad (11)$$

The nearest suitable standard value of $330 \text{ }\mu\text{F}$ was implemented. The coupling capacitors connected to the LED buffer stages were experimentally adjusted to $22 \text{ }\mu\text{F}$ to preserve the low-frequency control signal.

For the initial sizing of the coupling capacitors connected to the LED buffer stages, the input impedance of each buffer and the output impedance of the corresponding phase-splitter branch were taken into account. With $V_{BE} \approx 0.65 \text{ V}$, a DC bias current of approximately 1.30 mA , $h_{fe} = 140$, and $h_{ie} \approx 3 \text{ k}\Omega$, the equivalent resistance seen by each capacitor was estimated using the small-signal transistor model. Because the component parameters were approximate and waveform attenuation was observed at the lowest modulation rates, the final capacitor values were adjusted experimentally to $22 \text{ }\mu\text{F}$.

3 Audio Circuit

The audio path consists of a filtering stage, an LDR-based attenuation stage, and a passive summing network. Buffer stages biased from the same reference voltage isolate these sections from one another, reducing impedance interaction and improving voltage transfer between stages. The guitar signal is divided into low-pass and high-pass branches, independently modulated, and finally recombined at the output.

3.1 Buffer Stages and Biasing

The buffer stages were implemented using PNP transistors powered from a single 9 V supply. The nominal design target was to obtain an output swing of approximately 4 V peak at the final buffer stage. The initial operating-point estimates were:

- $I_{CQ} \approx 1.85 \text{ mA}$;
- $R_E = 2.7 \text{ k}\Omega$, assuming a $10 \text{ k}\Omega$ load;
- $V_{R_E} \approx 5 \text{ V}$;
- $V_{CEQ} \approx 4 \text{ V}$.

The small-signal estimates used $h_{fe} = 125$, $h_{ie} = 2.5 \text{ k}\Omega$, $V_{BE} \approx 0.65 \text{ V}$, and $h_{FE} = 128$.

The input impedance of each buffer can be approximated as

$$R_{in} = R_B \parallel [h_{ie} + h_{fe}(R_E \parallel R_L)]. \quad (12)$$

A value of $R_B = 150 \text{ k}\Omega$ was selected to limit the load placed on the preceding stage. In the final prototype, a common reference voltage was generated using a 330Ω and 120Ω voltage divider:

$$V_{ref} = 9 \text{ V} \frac{120 \Omega}{330 \Omega + 120 \Omega} = 2.4 \text{ V}. \quad (13)$$

The current through the divider is 20 mA . The final DC bias point of the buffers was adjusted experimentally; therefore, the nominal values presented above should be interpreted as initial design criteria rather than as a complete measurement of the final operating point.

3.2 Filters

The target crossover frequency for both branches was 800 Hz . Due to the selection of standard component values and experimental adjustments, the high-pass branch remained close to this value, while the low-pass branch exhibited a cutoff frequency of approximately 1.2 kHz . Detailed capacitor calculations for both branches are provided in Appendix A.

3.2.1 Low-Pass Branch

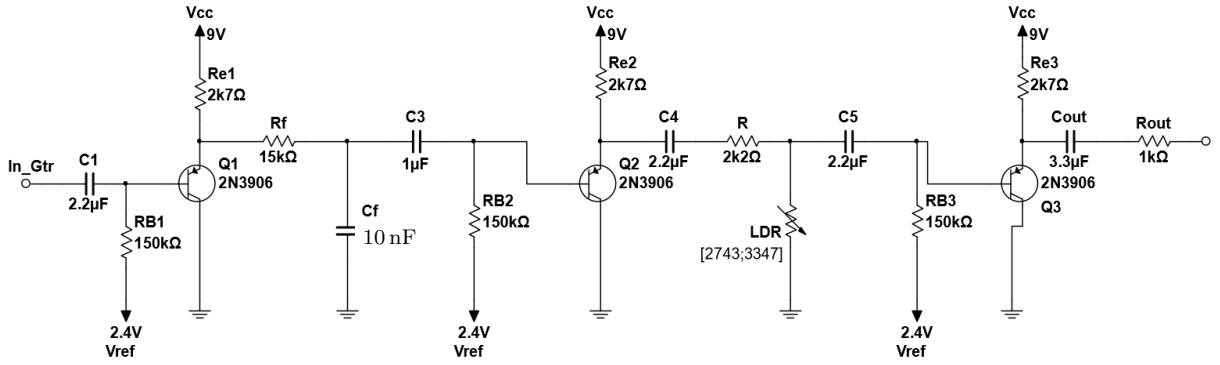


Figure 4: Low-frequency branch.

The coupling capacitors were designed for cutoff frequencies of 5 Hz , while the output capacitor was designed for 1 Hz . The main filter was initially calculated for a cutoff frequency of 800 Hz . The equivalent resistance seen by the filter capacitor was $12.62 \text{ k}\Omega$, resulting in

$$C_f = \frac{1}{2\pi(800 \text{ Hz})(12.62 \text{ k}\Omega)} = 15.8 \text{ nF}. \quad (14)$$

A standard value of 10 nF was implemented, producing

$$f_c = \frac{1}{2\pi(12.62 \text{ k}\Omega)(10 \text{ nF})} = 1.26 \text{ kHz}. \quad (15)$$

This value was experimentally approximated as 1.2 kHz. The implemented value of R_f was 15 k Ω , although the cutoff frequency depends on the total equivalent resistance seen by C_f .

3.2.2 High-Pass Branch

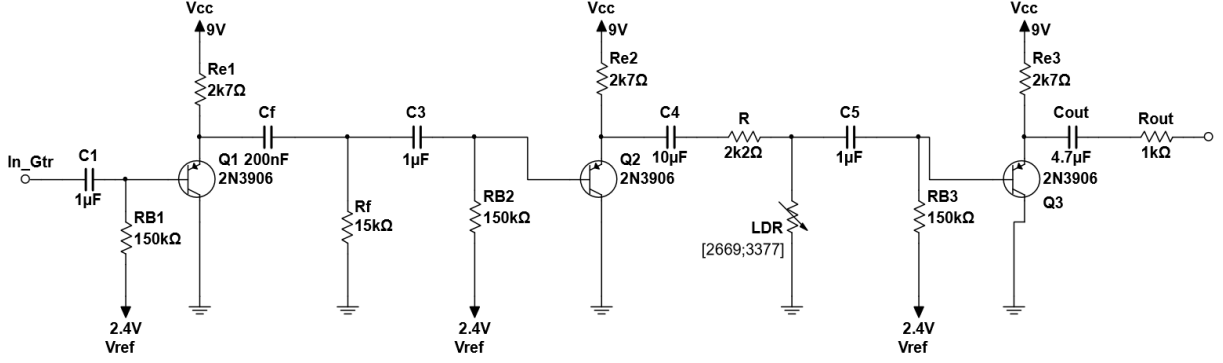


Figure 5: High-frequency branch.

The equivalent resistance of the high-pass filter was determined indirectly through experimental measurements and was approximately 1 k Ω . For a cutoff frequency of 800 Hz,

$$C_f = \frac{1}{2\pi(800 \text{ Hz})(1 \text{ k}\Omega)} = 199 \text{ nF}. \quad (16)$$

A standard value of 200 nF was implemented. The value of R_f in this branch was 15 k Ω , although this resistor alone does not represent the total equivalent resistance that determines the cutoff frequency.

3.3 Optical Gain Control

Each LED was optically coupled to an LDR that formed part of a voltage divider in its corresponding audio branch. A preliminary differential-pair configuration was used to characterize the LDR response before the final phase-splitter topology was adopted.

The LDR resistance was measured under DC conditions at several voltage levels representing instantaneous values of the control signal. The resulting data were fitted using a power-law model:

$$R_{LDR}(\Omega) = 761x^{-0.59}, \quad (17)$$

with a reported correlation coefficient of $r = -0.99$. Based on this characterization, a resistance of 2.2 k Ω was selected to produce a significant variation in the voltage-divider ratio.

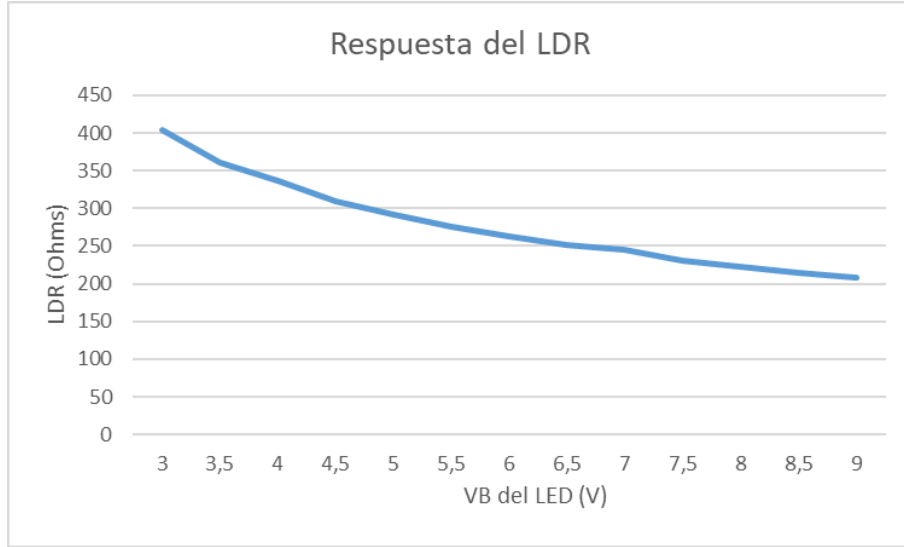


Figure 6: Experimental characterization of the LDR.

The resistance ranges measured in the prototype were:

Audio branch	Minimum resistance	Maximum resistance
Low-frequency branch	2743 Ω	3347 Ω
High-frequency branch	2669 Ω	3377 Ω

Table 1: Measured resistance range of the LDRs.

3.4 Measured Attenuation

The two branches were measured separately using a signal generator and an oscilloscope, before the passive summing stage. A 100 Hz test signal was used for the low-frequency branch. For the high-frequency branch, the original measurement record specifies 100 kHz. In both cases, the signal level was adjusted to prevent clipping in the buffer stages.

The decibel values shown in Table 2 were recalculated from the recorded linear amplitude ratios using $20 \log_{10}(A)$.

Operating condition	High-frequency branch	Low-frequency branch
Minimum of cycle, maximum depth	0.21 (−13.6 dB)	0.14 (−17.1 dB)
Maximum of cycle, maximum depth	0.95 (−0.4 dB)	0.90 (−0.9 dB)
Zero depth	0.35 (−9.1 dB)	0.30 (−10.5 dB)

Table 2: Measured amplitude ratio and attenuation in each audio branch.

The residual attenuation at zero depth occurs because both LEDs remain illuminated at their fixed DC bias point. Consequently, the LDRs retain a finite resistance within their respective voltage dividers.

4 Prototype Specifications

Parameter	Value
Power supply	9 V DC
Measured current consumption	200 mA
Input impedance	approximately 50 k Ω
Output impedance	approximately 1 k Ω
Modulation rate	1–10 Hz
Target crossover frequency	800 Hz
Implemented low-pass cutoff	approximately 1.2 kHz
Implemented high-pass cutoff	approximately 800 Hz
Controls	rate and depth

Table 3: Main specifications of the assembled prototype.

The impedance values are design estimates, while the current consumption was measured on the complete prototype. The cutoff frequencies correspond to the implementation built using the available components and should not be interpreted as production-level characterization.

5 Complete Circuit

5.1 Control Circuit

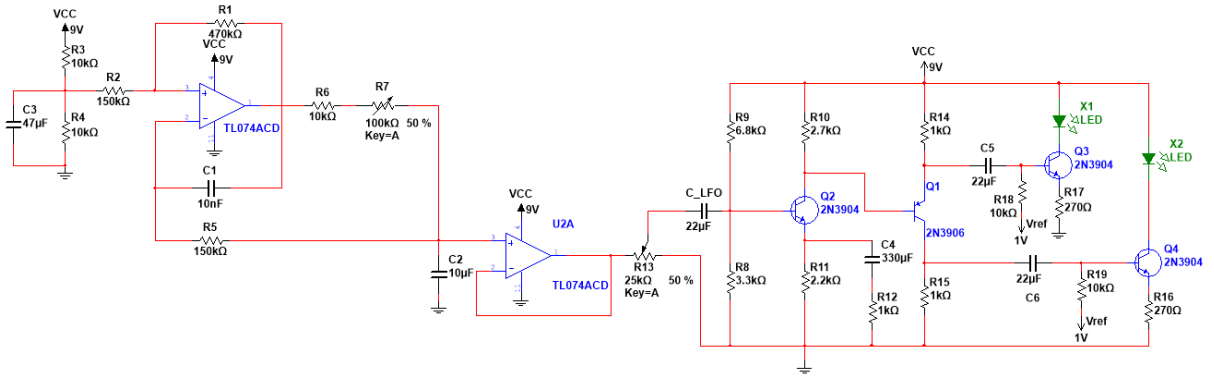


Figure 7: Complete schematic of the control circuit.

5.2 Audio Circuit

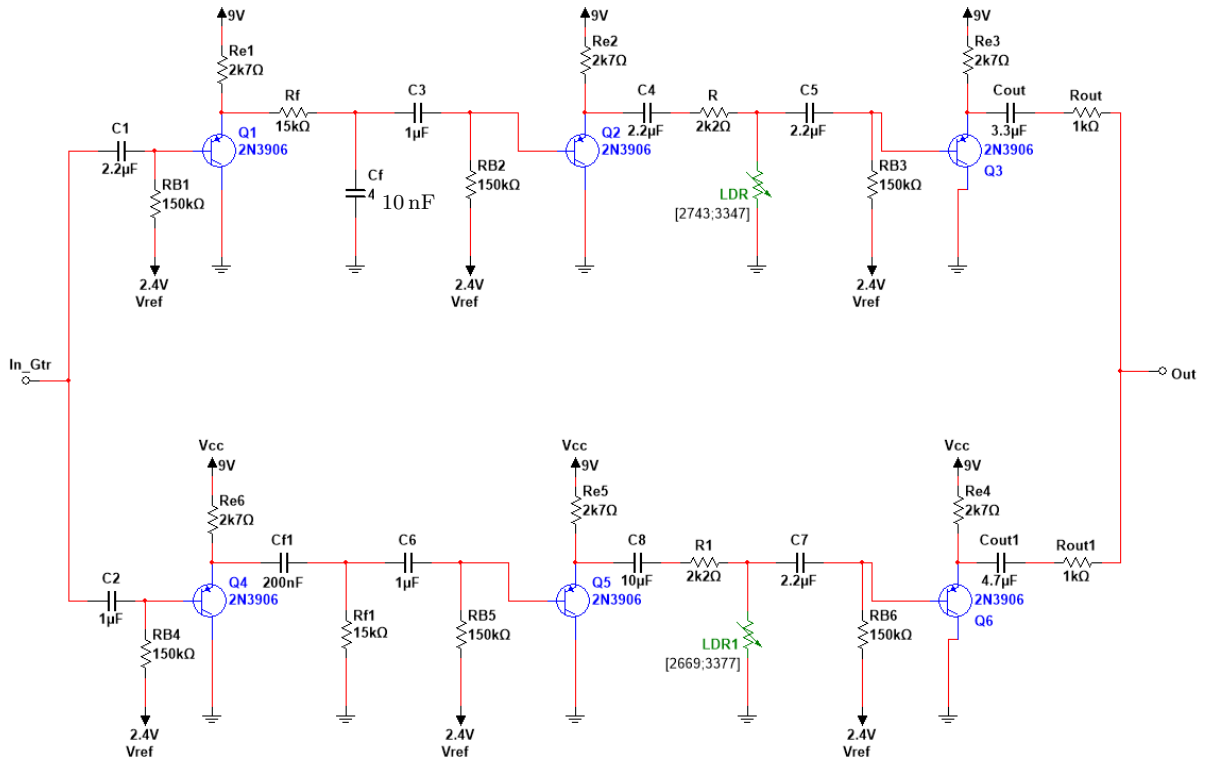


Figure 8: Complete schematic of the audio circuit.

6 Conclusions

An analog harmonic tremolo was designed and built to divide a guitar signal into two frequency bands and modulate them in opposite phase using two LED–LDR optocouplers. The rate and depth controls allowed the behavior of the effect to be adjusted without altering the DC bias point of the LEDs.

The measurements confirmed significant complementary modulation in both branches. At maximum depth, the measured attenuation ranged from approximately -13.6 to -0.4 dB in the high-frequency branch and from -17.1 to -0.9 dB in the low-frequency branch. At zero depth, the corresponding fixed attenuation values were approximately -9.1 dB and -10.5 dB.

The prototype also revealed several practical limitations. The main limitation was the difference between the intended 800 Hz low-pass cutoff frequency and the approximately 1.2 kHz cutoff obtained with the implemented standard component value. In addition, the inherent component variation of the LDRs produced small differences between the two audio branches. Future revisions should characterize the complete combined response, noise, distortion, and dynamic range, and could incorporate individual calibration of the optocouplers.

Despite these limitations, the project experimentally validated the operating principle and resulted in a functional device that was presented at CAPER 2024 and WorkShow 2024. The work integrated analog circuit design, small-signal analysis, transistor biasing, filtering, optoelectronics, laboratory measurement, and auditory evaluation.

A Audio-Circuit Capacitor Calculations

Equivalent-resistance diagrams were developed for the capacitor calculations using the following design criteria:

- The bypass capacitors were designed for a cutoff frequency of 5 Hz.
- The output coupling capacitors were designed for a cutoff frequency of 1 Hz.
- The main filter capacitor was designed for a cutoff frequency of 800 Hz.

Under these criteria, the equivalent resistance seen by each capacitor was calculated by treating capacitors with the same cutoff frequency as short circuits, capacitors with lower cutoff frequencies as short circuits, and capacitors with higher cutoff frequencies as open circuits.

Once the ideal capacitance values had been obtained, standard component values were selected. For high-pass sections, the capacitance was increased so that the actual cutoff frequency remained below the calculated value. For low-pass sections, the capacitance was reduced so that the actual cutoff frequency remained above the calculated value.

The worst-case LDR resistance was also considered. Its minimum resistance was used for the high-pass capacitor calculations, while its maximum resistance was used for the low-pass calculations.

Numerical verification confirmed that the ideal values presented below satisfy $C = 1/(2\pi f_c R_{eq})$ within the displayed rounding. The implemented standard values shift some auxiliary cutoff frequencies, but keep them below approximately 7 Hz and therefore well below the useful guitar-frequency range.

A.1 Low-Pass Circuit: Capacitors with a 5 Hz Cutoff

This section presents the calculations for the bypass capacitors in the audio branch designed to attenuate frequencies above 800 Hz.

A.1.1 C1

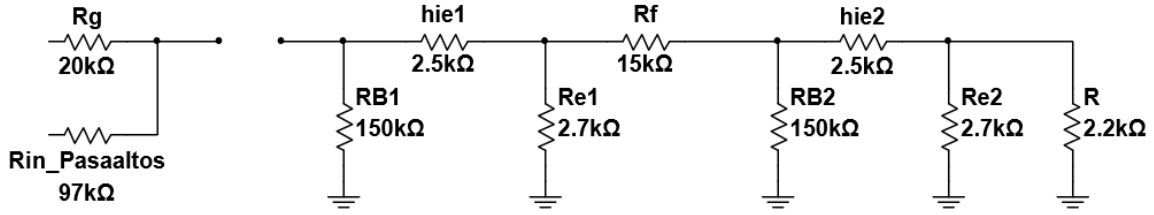


Figure 9: C1

$$R_{C1} = R_G // R_{in,HP} + (R_{B1} // (hie_1 + hfe_2 (Re_1 // (R_f + (R_{B2} // (hie_2 + hfe_3 (Re_2 // R))))))) \quad (18)$$

$$R_{C1} = 119.733 \text{ k}\Omega \quad (19)$$

$$C1 = 0.27 \mu\text{F} \quad (20)$$

A standard value of 2.2 μF was selected for this capacitor.

A.1.2 C3

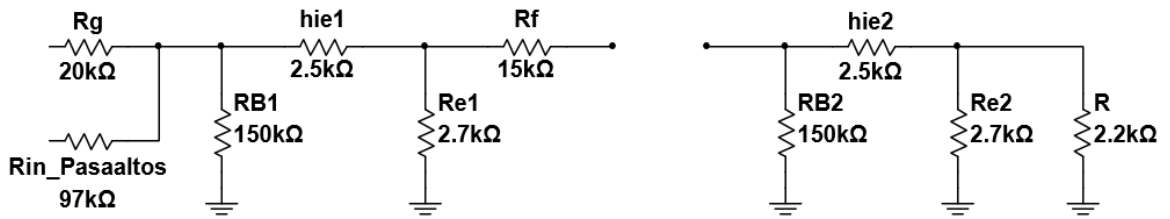


Figure 10: C3

$$R_{C3} = \left(\frac{(R_G // R_{In_{PA}} // R_{B1}) + hie_1}{hfe_1} // Re_1 \right) + R_f + (R_{B2} // (hie_2 + hfe_2 * (Re_2 // R))) \quad (21)$$

$$R_{C3} = 91.126 \text{ k}\Omega \quad (22)$$

$$C3 = 0.35 \mu\text{F} \quad (23)$$

A standard value of 1 μF was selected for this capacitor.

A.1.3 C4

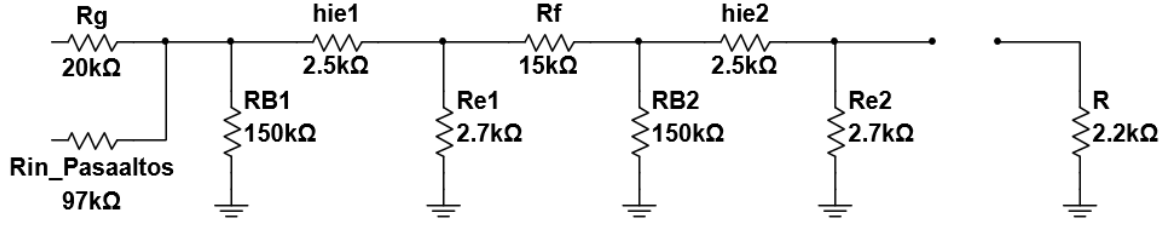


Figure 11: C4

$$R_{C4} = \frac{\left(\left(\left(\frac{R_G // R_{In_Pasaaltos} // R_{B1}}{hfe_1} \right) // R_{e1} \right) + R_f \right) // R_{B2} + hie_2}{hfe_2} // R_{e2} + R \quad (24)$$

$$R_{C4} = 2.324 \text{ k}\Omega \quad (25)$$

$$C4 = 13.7 \mu\text{F} \quad (26)$$

A standard value of $22 \mu\text{F}$ was selected for this capacitor.

A.1.4 C5

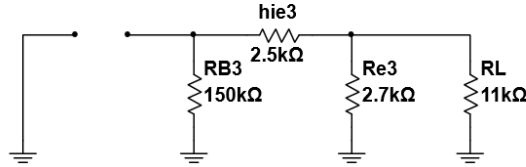


Figure 12: C5

$$R_{C5} = R_{In3} = R_{B3} // (hie_3 + hfe_3 * (Re_3 // R_L)) \quad (27)$$

$$R_{C5} = R_{In3} = 96.869 \text{ k}\Omega \quad (28)$$

$$C5 = 0.33 \mu\text{F} \quad (29)$$

A standard value of $2.2 \mu\text{F}$ was selected for this capacitor.

A.1.5 Cout

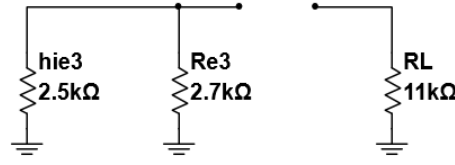


Figure 13: Cout

$$R_{COUT} = (\frac{hie_3}{hfe_3} // Re_3) + R_L \quad (30)$$

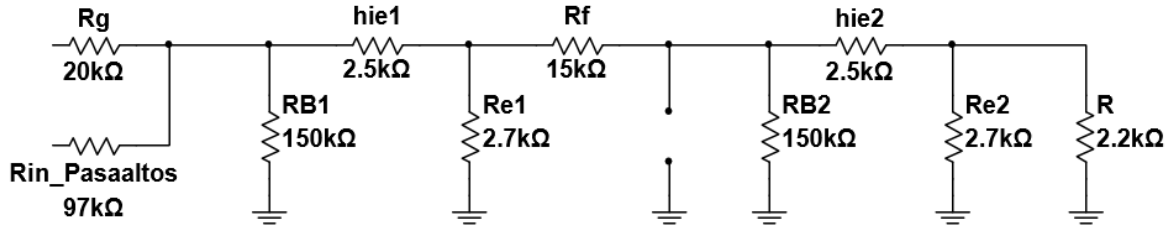
$$R_{COUT} = 11.020 \text{ k}\Omega \quad (31)$$

$$C_{OUT} = 2.9 \mu\text{F} \quad (32)$$

A standard value of $3.3 \mu\text{F}$ was selected for this capacitor.

A.2 Low-Pass Filter Capacitor for an 800 Hz Target Cutoff

This section presents the calculation of the capacitor used to attenuate frequencies above 800 Hz, so that the modulation is applied primarily below the cutoff frequency.

Figure 14: Low-pass filter capacitor, C_f .

$$R_{Cf} = (((\frac{R_G // R_{InPA} // R_{B1}}{hfe_1} + hie_1) // Re_1) + R_f) // (R_{B2} // (hie_2 + hfe_2 * (Re_2 // R))) \quad (33)$$

$$R_{Cf} = 12.620 \text{ k}\Omega \quad (34)$$

$$C_f = 15.8 \text{ nF} \quad (35)$$

A standard value of 10 nF was implemented, resulting in $f_c \approx 1.26 \text{ kHz}$, reported experimentally as approximately 1.2 kHz .

A.3 High-Pass Circuit: Capacitors with a 5 Hz Cutoff

For several of these calculations, the main filter capacitor, which has a higher cutoff frequency, is treated as an open circuit.

A.3.1 C1

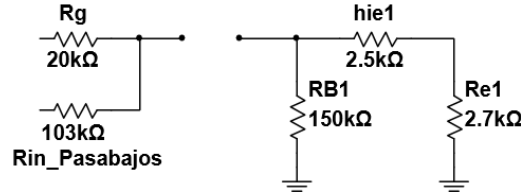


Figure 15: C1

$$R_{C1} = (R_G // R_{In_{PB}}) + R_{B1} // (h_{ie1} + h_{fe} * R_{e1}) \quad (36)$$

$$R_{C1} = 120.830 \text{ k}\Omega \quad (37)$$

$$C1 = 0.26 \mu\text{F} \quad (38)$$

A standard value of $1 \mu\text{F}$ was selected for this capacitor.

A.3.2 C3

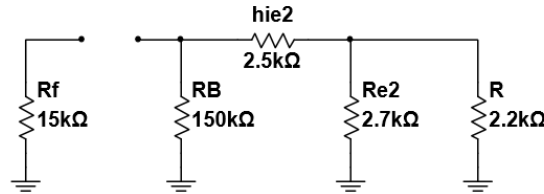


Figure 16: C3

$$R_{C3} = R_f + (R_{B2} // (h_{ie2} + h_{fe2} * (R_{e2} // R))) \quad (39)$$

$$R_{C3} = 90.994 \text{ k}\Omega \quad (40)$$

$$C3 = 0.35 \mu\text{F} \quad (41)$$

A standard value of $1 \mu\text{F}$ was selected for this capacitor.

A.3.3 C4

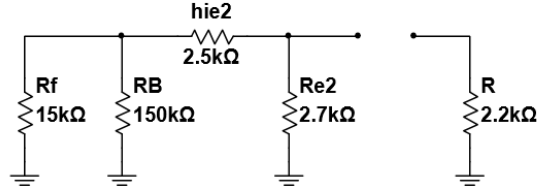


Figure 17: C4

$$R_{C4} = \left(\frac{R_f // R_{B2} + h_{ie2}}{h_{fe2}} // R_{e2} \right) + R \quad (42)$$

$$R_{C4} = 2.323 \text{ k}\Omega \quad (43)$$

$$C4 = 13.7 \mu\text{F} \quad (44)$$

A standard value of $10 \mu\text{F}$ was selected for this capacitor, corresponding to an approximate cutoff frequency of 6.9 Hz.

A.3.4 C5

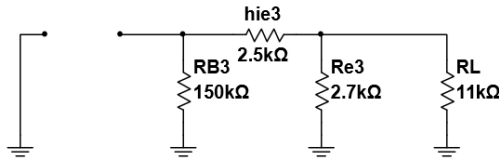


Figure 18: C5

$$R_{C5} = R_{B3} // (h_{ie3} + h_{fe3} * (R_{e3} // R_L)) \quad (45)$$

$$R_{C5} = 96.869 \text{ k}\Omega \quad (46)$$

$$C5 = 0.33 \mu\text{F} \quad (47)$$

A standard value of $1 \mu\text{F}$ was selected for this capacitor.

A.3.5 Cout

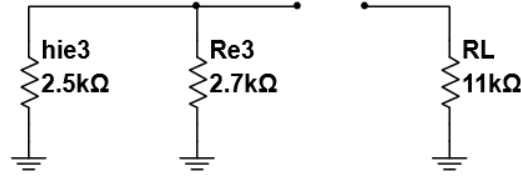


Figure 19: Cout

$$R_{C_{OUT}} = (\frac{hie_3}{h_{fe_3}} // Re_3) + R_L \quad (48)$$

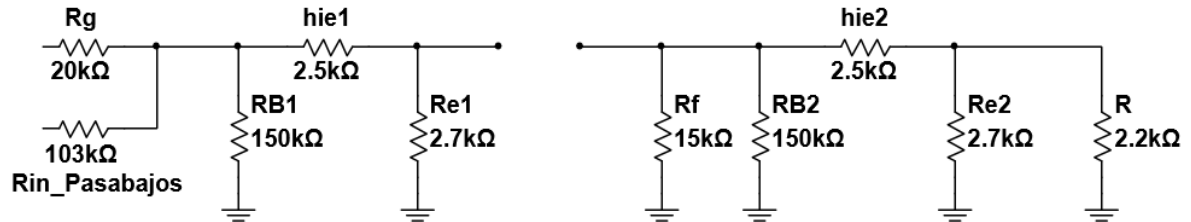
$$R_{C_{OUT}} = 11.020 \text{ k}\Omega \quad (49)$$

$$C_{OUT} = 2.9 \text{ }\mu\text{F} \quad (50)$$

A standard value of $4.7 \text{ }\mu\text{F}$ was selected for this capacitor.

A.4 High-Pass Filter Capacitor for an 800 Hz Target Cutoff

The equivalent resistance for this capacitor was initially calculated using the same procedure applied to the preceding capacitors and to the low-pass filter capacitor.

Figure 20: High-pass filter capacitor, C_f .

During experimental testing, however, the cutoff frequency was found to be above 800 Hz. A known capacitor was therefore installed and used to determine indirectly the equivalent resistance seen by the filter capacitor.

For this indirect measurement, a signal generator was connected to the audio circuit and the peak-to-peak output voltage was measured with an oscilloscope at a sufficiently high frequency, where the high-pass filter no longer produced significant attenuation. The measured reference voltage was then reduced by 3 dB to determine the expected peak-to-peak voltage at the cutoff frequency. The input frequency was gradually reduced until this voltage was reached. The measured frequency and the known capacitor value were then used to calculate the equivalent resistance:

$$R_{Cf} = \frac{1}{2\pi * f * C} = 1000\Omega \quad (51)$$

Using this equivalent resistance, the capacitor required for an 800 Hz high-pass cutoff was calculated as

$$C_f = \frac{1}{2\pi(800 \text{ Hz})(1000 \Omega)} = 199 \text{ nF}. \quad (52)$$

A standard value of 200 nF was implemented.